

Does a Fourth Depth Channel Improve YOLO Detection of Oil Palm Fresh Fruit Bunches? Results of 32 Experiments

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Abstract

This report answers one question. A YOLO detector reads three RGB channels. We added depth as a fourth channel. We measured whether detection improves, stays the same, or becomes worse.

The answer is that depth did not improve detection. Early fusion was harmful in two of three seeds. Middle and late fusion moved the score by less than the noise of the experiment. All twelve paired confidence intervals contain zero. A random-noise control channel moved the score by the same order of magnitude as a real depth channel.

We also measured whether the depth implementation itself was correct. It was not correct at the start. The sensor metadata claimed color alignment, but the buffer stayed in the depth-camera grid. We corrected this error with full reprojection. The negative result above comes from the corrected pipeline, not from the defective one.

Thirty-two numbered experiments (E-001 to E-032) and nine gates (G0 to G8) carry the evidence. This report lists every one of them with its result in one line.

Keywords: fresh-fruit-bunch detection, oil palm, RGB-D, four-channel input, Orbbec, YOLO26, fusion ablation

1. The question

A YOLO detector reads an image with three channels. These are red, green, and blue. A depth camera can supply a fourth channel. The fourth channel holds the distance from the camera to each pixel.

The idea is simple. Two fruit bunches can have the same color and can overlap in the image. The same two bunches can stand at different distances

from the camera. Depth can separate them when color cannot.

This report measures one thing:

- Does the fourth channel increase detection accuracy?
- Does it change nothing?
- Does it decrease accuracy because it adds noise?

The only way to answer this question is to run the experiment and to read the numbers. This report contains those numbers.

2. The answer

2.1. Short answer

The fourth channel did not increase detection accuracy in the tested regime. Two of three early-fusion seeds became worse. Middle fusion and late fusion moved the score less than the seed-to-seed variation of the same model.

Table 1: Result of the fusion matrix (E-032). Each value is the change in test mAP50 against the RGB-only model with the same seed.

Fusion point	Change per seed	Mean	Result
Early	−0.0120, +0.0234, −0.0017	+0.0032	Sign changes. No gain.
Middle	+0.0096, +0.0212, +0.0110	+0.0139	Positive in 3 of 3 seeds. Interval contains zero.
Late	−0.0056, +0.0070, +0.0102	+0.0039	Sign changes. No gain.
Random noise	−0.0130, +0.0025, −0.0081	−0.0062	Control arm. Carries no geometry.

All twelve paired confidence intervals contain zero. Middle fusion has the largest mean and a positive sign in all three seeds. This is an indication. It is not a confirmed improvement.

2.2. Why the noise arm matters

The noise arm is the most direct answer to the question. We replaced the depth channel with random numbers. The noise arm then moved the score by the same order of magnitude as the three depth arms. No contrast excluded zero.

This measurement does not prove that depth and noise are equivalent. We did not run an equivalence test. It shows that the depth values demonstrated no usable information in this regime, because a channel without geometry produced changes of the same size.

2.3. What the answer does not say

The answer applies to the tested regime. That regime is YOLO26n, 640 pixels, 150 epochs, one tree split, three seeds, and training from scratch. We did not prove that depth and RGB are equivalent. We did not measure a pre-trained middle-fusion model, because no compatible COCO checkpoint exists for the two-branch topology. We did not measure a larger palm plantation or a second camera model.

3. Was the depth implementation correct?

A negative result is only useful when the implementation is correct. We measured the implementation first. It contained five errors. We corrected all five before we read the final numbers. The first four errors belong to the depth pipeline and come from the E-022 audit. The fifth error is the evaluator asymmetry, which E-025 measured as a separate gate.

Table 2: Errors found in the depth pipeline, and their corrections (E-022 and E-025).

Error	Correction and measured effect
Wrong registration	The sidecar declared <code>alignedTo: color</code> , but the buffer stayed in the depth-camera grid. A direct resize missed the target by 29.3 pixels at the median and 61 pixels at the maximum. Full reprojection increased the mutual-information diagnostic by +0.0306 bits, with a 95 percent interval of [0.0260, 0.0354]. Mutual information is a registration diagnostic. It is not proof of geometric alignment, and it is not proof of a downstream benefit.
Shared calibration	Two camera units have different intrinsics. The pipeline now reads the calibration from the sidecar of each file.
Wrong depth range	The default range of 0.3 to 8.0 m did not fit this sensor. The E-022 pixel audit found no pixel below 0.3 m, and 10.07 percent of pixels above 8 m. The corrected train-only range is 0.8 to 15.0 m.
Cross-split donor and shared random state	The first favorable result used a donor image from another split and a shared random-number state. We withdrew that result.
Evaluator asymmetry	The two arms produced 2.44 times different detection density. This produced an offset of 0.0156, which is large enough to reverse the sign of the depth-minus-RGB difference. All later work uses one <code>pycocotools</code> paired evaluation.

The first depth result looked positive. That result came from the defective pipeline. Its own confidence interval already contained zero, at $[-0.0215, +0.0632]$. We withdrew it. Every number in Section 2 comes from the corrected pipeline.

3.1. Three kinds of depth, and why we measured all three

Table 3: The three depth sources in this study.

Source	Property	Why we measured it
Pseudo-depth	Predicted from the same RGB image. Relative scale.	It is free. If it works, no camera is necessary. Measured in E-003 to E-007.
Physical depth	Orbbec Y16 sensor. Metric millimeters.	It is an independent measurement. Measured in E-022 to E-032.
Random noise	Random values in the same tensor position.	It shows the cost of the extra channel alone. Measured in E-027, E-029, E-030, and E-032.

Pseudo-depth failed the direct measurement in E-006. The contrast of a bunch box was 0.0089 against 0.0341 for a size-matched random control. The difference in AUC was 0.009. Pseudo-depth comes from the same RGB image, so its error correlates with the RGB error. It is not an independent sensor.

Physical depth passed the file audit after the corrections in Table 2. It then failed the benefit measurement.

4. Evidence: all 32 experiments

Each row gives one experiment, its question, and its result. The full configuration, artifact path, and script name stay in the repository log at docs/experiments/EKSPERIMEN.md.

Table 4: The 32 numbered experiments and their results.

ID	Question	Result
E-001	Does the <code>class_mismatch</code> statistic measure maturity ambiguity?	No. 0 of 7,328 multi-view bunches disagree, because the same pipeline makes both the statistic and the link. FALSIFIED as a measure.
E-002	Can we reuse the raw master annotations directly?	No. The 3,992 raw files hold only 1,352 unique basenames, and 936 of those basenames repeat across folders. INCONCLUSIVE, so E-015 maps by content.
E-003	Can pseudo-depth organize one orbit video?	Partly. 48 frames cover 319.7 degrees with a circle residual of 8.2 percent. One video is not enough. PARTIAL.
E-004	Does that hold across six videos?	Yes for tree geometry. Five of six videos reach 270 degrees or more. One video reaches 41 percent. CONFIRMED for tree geometry only.
E-005	Can pseudo-depth order the photograph sides?	Yes. All 20 four-side trees and all 30 eight-side trees order correctly. Angular RMSE is 17.3 and 8.5 degrees against random baselines of 57.5 and 34.4 degrees. CONFIRMED.
E-006	Does pseudo-depth separate a bunch from its background?	No. Contrast is 0.0089 against 0.0341 for the control. The AUC difference is 0.009. FALSIFIED.
E-007	Can pseudo-depth link the same bunch across sides?	No. The simple statistical correction wins. The threshold sweep touched the test split, so the values are exploratory. FALSIFIED for this implementation.
E-008	—	Reserved number. No run. The slot stays visible so that a gap cannot look like a null result.
E-009	Is B4 hard because its boxes are small?	Not only. 81.2 percent of B4 boxes stay in the COCO medium band. The median box is 46 by 46 pixels. PARTIAL.

ID	Question	Result
E-010	Is B4 hard because bunches crowd each other?	No. B4 has the lowest neighbor count and the lowest maximum IoU, and low contrast. B2 has high ΔE and low AP50, so B2 is a different problem. Crowding FALSIFIED.
E-011	Can contrast or texture preprocessing recover B4?	Contrast boost fails. A Laplacian filter moves B4 from the least separated class to the most separated class, at AUC 0.6153. This supports a texture channel, not automatically a depth channel.
E-012	Is the maturity confusion random?	No. Accuracy is 52.87 percent against a 25 percent baseline, and the errors follow the chain B1–B2–B3–B4. The largest non-adjacent error is 1.9 percent. Maturity is ordinal.
E-013	What is the exact four-channel contract?	Channel order $[B, G, R, D]$, invalid depth is exact zero, the pipeline keeps a validity mask, and modality dropout keeps RGB-only inference possible. Contract complete. Device validation open.
E-014	Does the detector fail at localization or at maturity?	mAP50 is 0.7191 without maturity and 0.5218 with maturity, a ratio of 72.6 percent. PROVISIONAL, because we do not yet demonstrate byte-identical prediction JSON.
E-015	Can we map the master pixels safely?	Yes. 3,992 of 3,992 images map by content. The mapped set did not train E-021. UNBLOCKED.
E-016	Is 68 percent a hard maturity ceiling?	No. The three measurement paths share one crop classifier and use mismatched color augmentation. WITHDRAWN.
E-017	Does a two-stage detector help?	No. It is worse than the one-stage baseline on every class and on both mAPmetrics. FALSIFIED for this implementation.

ID	Question	Result
E-018	How far can localization reach at best?	Oracle reachability is 0.8834 and 0.4702. The median best IoU is 0.7303, and only 0.0376 of boxes pass IoU 0.90. This explains why mAP50-95 stays hard. Descriptive envelope.
E-019	Do more pixels close the gap?	No. A single color-safe high-resolution run stops at epoch 41 and does not reach the target. We remove resolution alone as the next step.
E-020	Does an NMS-free transformer help?	RT-DETR-L improves all four classes, with the largest gain on B4. Many factors change together, so this is a shortlist result.
E-021	Which RGB detector is the fair control?	RF-DETR-L. Test mAP50 is 0.6038 and test mAP50-95 is 0.2770. The paired difference against RT-DETR-L is +0.0255, with a 95 percent interval of [0.0104, 0.0408]. Best observed RGB benchmark.
E-022	Do the physical depth files measure what the metadata claims?	No. See Table 2. The registration, range, donor, and random-state errors are corrected. The first positive delta is WITHDRAWN.
E-023	—	Execution alias. The redesigned matrix ran once, as E-032. It is not a second measurement.
E-024	Can we measure cross-side identity without a circular proxy?	Yes, from the annotation graph. Collisions concentrate in B1–B2 and B2–B3, with no B1–B3 collision. B4 has zero two-side detections, so power is limited.
E-025	Do the trainer metric and <code>pycocotools</code> agree?	No. The two arms differ in detection density by 2.44 times, which creates an offset of 0.0156. That offset is large enough to reverse the sign. One evaluator now binds all later work.

ID	Question	Result
E-026	Does depth stabilize cross-side identity?	INCONCLUSIVE. The denominators are 82 RGB bunches and 75 RGB-D bunches, so the comparison is not fully paired. The interval is too wide.
E-027	Does early fusion help across three seeds?	No. Two of three intervals are negative, and only seed 42 is positive. The benefit criterion is FALSIFIED for YOLO26n.
E-028	Does a larger dataset change the identity measure?	It adds power, not a difference. The rate is 0.2329 against 0.1951, a difference of +0.0378 with an interval of $[-0.0585, +0.1276]$. No physical depth is present in this dataset.
E-029	Does a larger model rescue depth?	No. The depth-minus-noise mean is +0.0124, below the archived threshold of +0.015. The RGB arm alone varies by 0.0759 across seeds. The capacity-rescue clause is RETRACTED.
E-030	Does the effect depend on model capacity?	Unclear. Noise-minus-RGB is +0.0032, +0.0184, -0.0325 , -0.0533 . The sign crosses once, between 21.9 and 26.3 million parameters, but the magnitudes are not monotonic. One seed only. This opens G7b.
E-031	Is the split effect larger than the seed effect?	Yes for absolute scores. The absolute mAPspans 0.0488 across splits against 0.0321 across seeds. Every absolute score must name its split.
E-032	Is the fusion point the explanation?	INCONCLUSIVE. Five arms, three seeds, twelve paired intervals, and all twelve contain zero. See Table 1.

4.1. How the 32 records answer the one question

Six records give the direct answer. These are E-022, E-027, E-029, E-030, E-031, and E-032. They train the detector with and without the fourth channel and compare the scores.

The other records remove the alternative explanations for that answer. Without them, a reader can reject the negative result for a reason that has nothing to do with depth:

- E-001, E-002, E-015, and E-024 make sure that the labels and the identity graph are not circular.
- E-025 makes sure that both arms use one evaluator. Before this record, the evaluator alone can reverse the sign.
- E-031 makes sure that the split is named. The split moves the absolute score more than the seed does.
- E-003 to E-007 remove pseudo-depth as a cheap substitute for the sensor.
- E-009 to E-012 explain what the RGB error actually is. B4 is a contrast problem and B2–B3 is an ordinal maturity problem. Depth can help the first one. Depth cannot help the second one.
- E-013 freezes the channel contract, so that a later run cannot change the channel order in silence.
- E-017 to E-021 supply a strong RGB control. A weak control makes any fourth channel look useful.

5. Gate register G0 to G8

The depth work ran as nine gates. A gate is a hole that must close before we accept the conclusion. One later gate, G7b, is still open.

Table 5: Gate register. G0, G4, G6, and G7b are not closed.

Gate	Content	Closed by	Status
G0	Run completeness and dead links	d58eae9	Open. We must release the manifest and the provenance with the source and the PDF.
G1	Difference between the trainer metric and <code>pycocotools</code>	E-025	Asymmetry bound. The full mechanism is not resolved.
G2	Multi-seed matrix under one protocol	E-027, E-029	Closed for the tested regime. Not universal.
G3	E-022 restructure and SR-015 alignment	9c5d9dd, 86f8e65	Closed.
G4	Middle fusion	E-032	Inconclusive. The intervals contain zero. E-032 does not prove equivalence.
G5	Split variance	E-031	Measured on three splits. No population law.
G6	Late fusion	E-032	Inconclusive. The intervals contain zero. E-032 does not prove equivalence.
G7	Capacity sweep in one family	E-030	Exploratory. One seed. The capacity claim stays open.
G8	Cross-side measure on a dataset with usable power	E-028	RGB power increased. This is not a depth comparison.
G7b	Capacity monotonicity across seeds	—	Open.

5.1. The state of G7b

Commit 7afd274 opened 12 additional runs. These runs cover yolo26m and yolo26l, three modalities, and the seeds 1337 and 2024. The true state is:

- 7 of 12 runs finished training, and their curves are archived.
- We computed no paired contrasts. The paired files exist only for seed 42.
- We never updated E-030, so its one-seed limitation still applies.

The capacity turning point between 21.9 and 26.3 million parameters is therefore a one-seed pattern. It is not a finding. Five runs and one paired evaluation close this gate.

6. What this means for the field system

- Use the RGB path. RF-DETR-L is the best observed control, at test mAP50 0.6038. Its speed is 8.5 FPS on an L4 GPU, which is too slow for real-time field use without optimization.
- Do not add the depth channel to raise accuracy. The measurement does not support that decision.
- If depth hardware is already present, keep the capture as an auditable data branch. Apply the per-file quality gate from Table 2.
- For the B2–B3 maturity error, use an ordinal loss or a regression head. E-012 points there. A fourth channel does not address an ordinal boundary.
- The one-count-per-bunch inventory target is not validated. No experiment in this report measures it.

7. Limits of this answer

- The fusion matrix used YOLO26n only, at 640 pixels, for 150 epochs, on one split, and from scratch.
- No pretrained checkpoint exists for the two-branch topology, so transfer learning was not measured.
- Three seeds and three splits expose variance. They do not estimate a population distribution.
- The depth dataset holds 352 trees. The RGB pilot holds 953 trees.
- The claims apply to the audited Orbbec Y16 files. They do not apply to every Orbbec camera or every depth encoding.
- We never trained RF-DETR-L with physical depth.

8. Conclusion

We added a fourth depth channel to a YOLO detector and measured the effect. In the tested regime, the channel did not improve detection. Early fusion was harmful in two of three seeds. Middle and late fusion produced changes that a random-noise channel also produced. All twelve paired intervals contain zero.

The depth implementation was defective at the start, and we corrected it. The final answer comes from the corrected pipeline, so the negative result belongs to the sensor and the method, not to a registration error.

The answer is conditional. Section 7 lists the conditions. One gate, G7b, can still change the capacity part of the answer. It needs five runs and one paired evaluation.